

Compact Hall Geometry and Finite-Height Positive Borchers Recognition

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Abstract

Extensions of coherent sheaves provide the geometric input for a compact Hall algebra. We ask when specified classes in such an algebra realize the positive part of a Borchers algebra. We first describe the extension correspondence, its critical coefficients, and the common translation torsor required for sheaves on a K3 surface times an elliptic curve. The translation quotient has zero-object carrier BE . We then prove finite-height recognition for a fixed, purely even Borchers presentation in symmetric monoidal height truncations. Relations construct the canonical enveloping-to-generated-Hall surjection. Homogeneous dimensions can prove that it is an isomorphism. An independently constructed Hall-to-enveloping quotient gives another criterion and determines the additional kernel in the full Hall algebra. Compatible comparisons extend to the specified height completion. Scalar denominators alone determine none of these maps. Reduced critical coefficients, coherent orientations, and the compact bialgebra comparison remain necessary for the geometric realization.

1 Extension correspondences and critical coefficients

An extension combines two sheaves while retaining how they are joined. A Hall product must carry this extension data into its coefficients. The comparison with a Borchers algebra therefore starts with the moduli stack and its extension correspondence.

Throughout, $\mathbb{k}$ is a field of characteristic zero. The geometric constructions take place over $\mathbb{C}$. Fix a finite set I and the charge lattice

$$Q = \bigoplus_{i \in I} \mathbb{Z}\alpha_i, \quad Q_+ = \bigoplus_{i \in I} \mathbb{Z}_{\geq 0}\alpha_i.$$

Let X be a smooth projective complex Calabi–Yau threefold. Choose an extension-closed subcategory $\mathcal{A} \subset \text{Coh}(X)$, closed under direct sums, whose objects form an open derived substack of the derived moduli stack of perfect complexes on X . Choose a charge map

$$\text{cl} : K_0(\mathcal{A}) \longrightarrow Q$$

which is locally constant in families and whose effective classes lie in Q_+ . Assume that the Euler form factors through an antisymmetric bilinear form $\chi : Q \times Q \rightarrow \mathbb{Z}$:

$$\chi(\text{cl } E, \text{cl } F) = \sum_{r \in \mathbb{Z}} (-1)^r \dim_{\mathbb{C}} \text{Ext}^r(E, F).$$

Let $\mathfrak{M}_\beta$ be the open-and-closed derived substack of objects of charge β . With the convention that the first factor is the subobject, the stack $\mathfrak{E}_{\beta,\gamma}$ of short exact sequences

$$0 \longrightarrow E_\beta \longrightarrow E \longrightarrow E_\gamma \longrightarrow 0$$

has the typed correspondence

$$\mathfrak{M}_\beta \times \mathfrak{M}_\gamma \xleftarrow{p_{\beta,\gamma}} \mathfrak{E}_{\beta,\gamma} \xrightarrow{q_{\beta,\gamma}} \mathfrak{M}_{\beta+\gamma}. \quad (1)$$

The (-1) -shifted symplectic structure supplies, after a choice of orientation o , Donaldson–Thomas coefficient complexes $\mathcal{P}_{\beta,o}$. A prospective cohomological carrier is

$$\mathcal{H}_\beta = \mathbb{H}^\bullet(\mathfrak{M}_\beta, \mathcal{P}_{\beta,o}), \quad \mathcal{H} = \bigoplus_{\beta \in Q_+} \mathcal{H}_\beta. \quad (2)$$

For a moduli-like stack whose classical truncation is a quasi-separated Artin stack with affine stabilizers and is Θ -reductive, and for which the unit inclusion is closed, a *strong orientation datum* contains an orientation isomorphism at the zero object and its two unit coherences, as well as the compatibility with direct sums and the coherence for triple direct sums. It yields maps of the form

$$m_{\beta,\gamma} : \mathcal{H}_\beta \otimes \mathcal{H}_\gamma \longrightarrow \mathcal{H}_{\beta+\gamma}[\chi(\beta,\gamma)], \quad (3)$$

where χ is the antisymmetric Euler pairing. The shift in (3) is part of the structure: componentwise shifts cannot remove a nonzero antisymmetric bilinear degree. Indeed, replacing $\mathcal{H}_\beta$ by $\mathcal{H}_\beta[s(\beta)]$ changes the degree to

$$\chi_s(\beta,\gamma) = \chi(\beta,\gamma) + s(\beta) + s(\gamma) - s(\beta+\gamma), \quad \chi_s(\beta,\gamma) - \chi_s(\gamma,\beta) = 2\chi(\beta,\gamma).$$

Thus the product must be placed in its specified charge-twisted monoidal category.

Kontsevich and Soibelman construct this structure for quivers with potential [6, §7]. In the general three-Calabi–Yau setting, the hypotheses above and strong orientation data give associativity by [5, Theorem A, Definition 8.4, Corollary 8.8]. Joyce and Upmeyer construct canonical orientation data for smooth projective Calabi–Yau threefolds [4, Theorem 4.4]. They identify the obstruction to a strong lift as a class in $H^3(K_0^{\text{semi}}(X), \mathbb{Z}/2)$ and leave its vanishing open [4, Theorem 3.10, Question 3.11]. Without that coherence, the product is associative only up to choices of sign [5, Remark 8.7]; the compactly supported construction of [3, Theorem 7.22] likewise assumes a strong orientation datum and records the unresolved lift in [3, Remark 7.23]. Thus (1) and (2) do not yet furnish the required associative compact Hall algebra.

Remark 1.1. *For the two-Calabi–Yau category of semistable coherent sheaves of fixed slope on a projective K3 surface, the ψ -sign-twisted BPS algebra is an enveloping algebra of a generalized Kac–Moody Lie algebra [2, §6.3, Theorems 1.1 and 1.2]. This is a compact two-Calabi–Yau theorem with graded generator spaces. The BPS algebra embeds in the full cohomological Hall algebra [2, Proposition 6.4], and twisting both multiplications preserves this inclusion. After choosing a positive determinant line bundle, the PBW comparisons for the full algebra are not algebra morphisms [2, Theorem 1.7, Corollary 1.8, and the following paragraph]; they therefore do not supply the three-Calabi–Yau algebra or bialgebra surjection required in Proposition 5.2.*

1.1 Translation quotients retain one torsor

For $X = S \times E$, where S is a projective K3 surface and E is an elliptic curve, translation changes the elliptic origin. An extension still depends on the relative positions of its endpoint sheaves. Quotienting the two origins independently loses this information.

Work in the fppf topology, with 2-fibre products of stacks. The stack BE classifies right E -torsors. For a stack M with a left E -action, an object of $[M/E](T)$ is a right torsor $P \rightarrow T$ and an equivariant map $m : P \rightarrow M$, with $m(pe) = e^{-1}m(p)$. For stack-valued maps this description includes their descent isomorphisms. Forgetting m gives $[M/E] \rightarrow BE$.

Proposition 1.2. *For E -stacks M, N , there is a natural equivalence*

$$[(M \times N)/E_{\text{diag}}] \simeq [M/E] \times_{BE} [N/E].$$

These equivalences commute with regrouping any finite number of factors.

Proof. An object on the right consists of (P, m) , (Q, n) , and a torsor isomorphism $\alpha : P \rightarrow Q$. Send it to $(P, m, n\alpha)$. Conversely, use the same torsor twice and its identity isomorphism. For a morphism, the relation $b\alpha = \alpha'a$ determines the second torsor map b from the first map a . The constructions are inverse up to their specified isomorphisms and commute with base change. For several factors, every regrouping gives one torsor with the same equivariant maps. This also proves the regrouping coherence. $\square$

For example, translation gives $[E/E] \simeq \text{Spec } \mathbb{C}$, whereas

$$[(E \times E)/E_{\text{diag}}] \simeq E, \quad (x, y) \longmapsto y - x.$$

Thus the fibre product over BE retains the relative translation. The ordinary product of the two quotient stacks is a point.

Let $\mathfrak{M}$ be the stack of coherent sheaves on X , with families flat over the parameter scheme. Put $\mathfrak{Q} = [\mathfrak{M}/E]$. An object over T is a torsor $P \rightarrow T$ and a flat coherent sheaf on $X_P = P \times^E X \simeq S \times P$. Choose an ample line bundle on X . Hilbert polynomials are invariant under the connected translation group, so they define substacks $\mathfrak{Q}_a$ independently of the local trivialization of P .

Let $\mathfrak{H}_{a,b}$ classify a common torsor and a sequence

$$0 \longrightarrow A \longrightarrow F \longrightarrow B \longrightarrow 0$$

on X_P , with flat terms of Hilbert polynomials $a, a+b, b$. The correspondence is

$$\mathfrak{Q}_a \times_{BE} \mathfrak{Q}_b \xleftarrow{p} \mathfrak{H}_{a,b} \xrightarrow{q} \mathfrak{Q}_{a+b}. \quad (4)$$

Here p records the endpoint sheaves on their common torsor, and q records the middle sheaf.

Proposition 1.3. *The correspondences (4) define a coherently associative unital Hall object in the category of spans of stacks over BE , with tensor product $\times_{BE}$. Its unit is*

$$BE \xleftarrow{\text{id}} BE \xrightarrow{0} \mathfrak{Q}.$$

Proof. Both triple products classify a common torsor and a flag $0 \subset A \subset D \subset F$. For the bracketing that first forms F/A , recover D as the inverse image in F of the prescribed subobject of F/A . Conversely, a flag gives both pairs of short exact sequences. These constructions preserve the endpoint sheaves, the middle sheaf, and base change. They identify the two composite spans.

For any number of factors, every parenthesization reconstructs the same full flag by inclusions and inverse images. The resulting identifications preserve each subobject and specified quotient, so their composites agree. This gives the associativity coherences. The zero sheaf has no sheaf automorphisms, but retains its translation torsor. Its quotient is therefore BE . Inserting a zero step in a flag gives the two unit identities and their coherences. $\square$

An E -invariant extension-closed sector inherits this construction when its families and admissible sequences satisfy descent and base change. Properness of the restricted map q remains a separate condition.

A cohomological realization of (4) must specify coefficients, the pullback or Gysin map along p , pushforward along q , their shifts, and coherent base change on the flag diagrams. Its tensor comparison must realize the fibre product over BE . Applying such a monoidal realization to Proposition 1.3 gives an associative unital algebra in its target category.

The zero-object stabilizer is the elliptic curve E , which is not affine. Consequently, the affine-stabilizer hypothesis of the absolute critical Hall theorem cited above fails for $\mathfrak{Q}$. The geometric flag construction does not supply a relative critical coefficient theory over BE . Removing translation or trace directions also requires a reduced coefficient complex, with its own duality and extension comparisons.

1.2 The orientation cross line

Choose an E -invariant holomorphic volume form on $X = S \times E$. For perfect families on $\pi : X_P \rightarrow T$, suppose the complexes

$$C(A, B) = R\pi_* \mathrm{RHom}_{X_P}(A, B)$$

are perfect over T . Write $L(A, B) = \det C(A, B)$ and $D(A) = L(A, A)$. The determinant convention is $\det C = \bigotimes_i (\det C^i)^{(-1)^i}$, with parity $\chi(C) \bmod 2$.

Relative Serre duality gives $C(B, A) \simeq C(A, B)^\vee[-3]$. Duality and the odd shift each invert the determinant line, hence $L(B, A) \simeq L(A, B)$. For an extension $0 \rightarrow A \rightarrow F \rightarrow B \rightarrow 0$, determinant additivity in the two variables gives

$$\begin{aligned} D(F) &\simeq L(B, A) \otimes D(A) \otimes D(B) \otimes L(A, B) \\ &\simeq D(A) \otimes D(B) \otimes L(A, B)^{\otimes 2}. \end{aligned}$$

The second line uses the determinant symmetry, including its parity sign. This is the extension identity of [4, equations (4.3)–(4.5)].

An orientation line o_A is a square root of $D(A)$. Extension compatibility requires an isomorphism

$$o_A \otimes o_B \otimes L(A, B) \longrightarrow o_F \tag{5}$$

whose square is the determinant comparison. For three successive quotients, both bracketings have cross factors $L(A, B) \otimes L(A, C) \otimes L(B, C)$. Agreement of their squares leaves a possible

μ_2 discrepancy. Strong orientation data require agreement of the square-root maps themselves, together with unit coherence. On the quotient they also require compatible E -linearizations. The displayed calculation concerns unreduced complexes. It does not construct these data for a reduced critical theory.

The correspondences specify how a Hall multiplication must arise. Recognition of a Borchers algebra then requires actual classes and relations in that multiplication. The following example shows why its graded dimensions cannot replace those operations.

2 Scalar denominators and Lie brackets

For a positively graded vector space $V = \bigoplus_{n \geq 1} V_n$ with finite-dimensional components, write

$$D_V(t) = \prod_{n \geq 1} (1 - t^n)^{\dim_{\mathbb{K}} V_n}.$$

Proposition 2.1 (The scalar shadow is not faithful). *There are nonisomorphic positively graded Lie algebras L and L_0 for which*

$$D_L(t) = D_{L_0}(t) \quad \text{and} \quad \text{Hilb}_t U(L) = \text{Hilb}_t U(L_0).$$

Proof. Let L have basis x, y in degree one and z in degree two, with $[x, y] = z$ and z central. Let L_0 have the same graded vector space and zero bracket. Then

$$[L, L] = \mathbb{K}z, \quad [L_0, L_0] = 0,$$

so the two Lie algebras are not isomorphic. Their common scalar denominator is

$$D_L(t) = D_{L_0}(t) = (1 - t)^2(1 - t^2).$$

The Poincaré–Birkhoff–Witt theorem gives

$$\text{Hilb}_t U(L) = \text{Hilb}_t U(L_0) = \frac{1}{(1 - t)^2(1 - t^2)}.$$

Nevertheless, $U(L_0)$ is commutative, whereas $xy - yx = z \neq 0$ in $U(L)$. □

Thus graded dimensions determine neither the Lie bracket nor the associative product of its enveloping algebra. They also do not determine the correspondences, orientations, or functorial maps from which a Hall product is formed.

3 The positive Borchers presentation

A comparison on generators requires a presentation of the target Lie algebra. Use the finite index set I of the charge lattice. An *integral symmetrizable Borchers–Cartan matrix* is an integral matrix $A = (a_{ij})_{i,j \in I}$ such that

1. $a_{ii} = 2$ or $a_{ii} \leq 0$;
2. $a_{ij} \leq 0$ for $i \neq j$, and $a_{ij} = 0$ if and only if $a_{ji} = 0$;

3. there are positive integers d_i for which $d_i a_{ij} = d_j a_{ji}$.

Put

$$I^{\text{re}} = \{i \in I : a_{ii} = 2\}, \quad I^{\text{im}} = \{i \in I : a_{ii} \leq 0\}.$$

The symmetric form on $Q = \bigoplus_{i \in I} \mathbb{Z} \alpha_i$ is $(\alpha_i, \alpha_j) = d_i a_{ij}$. Thus the simple roots indexed by I^{re} have positive square and those indexed by I^{im} have nonpositive square. We work in ordinary vector spaces: every simple generator is even, and no Lie-superalgebra signs occur.

Write $Q_+ = \bigoplus_{i \in I} \mathbb{Z}_{\geq 0} \alpha_i$ and

$$\text{ht} \left(\sum_i n_i \alpha_i \right) = \sum_i n_i.$$

Let $\mathfrak{F}$ be the free Q_+ -graded Lie algebra on generators e_i of degree α_i . Let $R_A \subset \mathfrak{F}$ be the homogeneous ideal generated by

$$(\text{ad } e_i)^{1-a_{ij}} e_j = 0 \quad (i \in I^{\text{re}}, j \neq i), \quad (6)$$

$$[e_i, e_j] = 0 \quad (i \neq j, a_{ij} = 0). \quad (7)$$

The positive Borchers Lie algebra considered here is

$$\mathfrak{n}_+(A) = \mathfrak{F}/R_A.$$

These are the normalized positive relations in Borchers' presentation [1, §1]. A simple imaginary generator e_i , $i \in I^{\text{im}}$, imposes no relation $(\text{ad } e_i)^m e_j = 0$ unless the orthogonality relation (7) applies. A relation based at a real index j may still constrain e_i . For example, for

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 0 \end{pmatrix}$$

the relation based at 1 is $(\text{ad } e_1)^2 e_2 = 0$, and there is no converse Serre relation based at 2.

Finiteness of I implies that every Q_+ -homogeneous component of $\mathfrak{F}$, and hence of $\mathfrak{n}_+(A)$, is finite-dimensional.

Remark 3.1 (What the positive presentation does not recover). *The matrix A is fixed input. Its positive Lie algebra need not determine A : the rank-one matrices (0) and (-2) both give $\mathfrak{n}_+ = \mathbb{k}e$. Recognition of $\mathfrak{n}_+(A)$ therefore does not recover the Cartan matrix, the Cartan and negative summands, or the full Borchers algebra. The present formulation also assumes a finite set of independent simple charges, one even generator for each charge. Charged generator spaces, odd simple roots, and Borchers–Bozec generators in multiple degrees are outside its scope.*

4 Finite-height recognition

A finite charge calculation retains products only when their total height stays within the chosen bound. To retain the coproduct as well, the tensor product must impose the same total-height condition.

For $N \geq 1$, let $\mathbf{Gr}_{\leq N}(Q_+)$ be the category of Q_+ -graded $\mathbb{k}$ -vector spaces supported in heights at most N . Its tensor product is

$$(V \bar{\otimes}_N W)_\beta = \bigoplus_{\gamma+\delta=\beta} V_\gamma \otimes W_\delta, \quad \text{ht}(\beta) \leq N.$$

With the usual symmetry this is a symmetric monoidal category. The functor

$$\tau_{\leq N} V = \bigoplus_{\text{ht}(\beta) \leq N} V_\beta$$

from nonnegatively Q_+ -graded vector spaces to $\mathbf{Gr}_{\leq N}(Q_+)$ is strong symmetric monoidal. Indeed, if $\beta = \gamma + \delta$ and $\text{ht}(\beta) \leq N$, then both γ and δ have height at most N . It therefore carries Lie algebras, associative algebras, coalgebras, and bialgebras to objects of the same kind. For $M \leq N$,

$$\tau_{\leq M} \tau_{\leq N} = \tau_{\leq M}. \quad (8)$$

In particular, $\tau_{\leq N} U(\mathfrak{g})$ is a bialgebra object in the truncated category. The ordinary algebra quotient obtained by killing heights above N need not be an ordinary coalgebra quotient.

Theorem 4.1 (Finite-height Lie recognition). *Let*

$$L = \bigoplus_{\beta \in Q_+ \setminus \{0\}} L_\beta$$

be a Q_+ -graded Lie algebra with finite-dimensional homogeneous pieces. Suppose that elements $x_i \in L_{\alpha_i}$ generate L and satisfy (6) and (7), with e_i replaced by x_i . Then there is a unique surjective graded Lie homomorphism

$$\pi : \mathfrak{n}_+(A) \longrightarrow L, \quad e_i \longmapsto x_i. \quad (9)$$

If, for every $\beta \in Q_+ \setminus \{0\}$ with $\text{ht}(\beta) \leq N$,

$$\dim_{\mathbb{k}} L_\beta = \dim_{\mathbb{k}} \mathfrak{n}_+(A)_\beta, \quad (10)$$

then

$$\tau_{\leq N} \pi : \tau_{\leq N} \mathfrak{n}_+(A) \xrightarrow{\sim} \tau_{\leq N} L$$

is an isomorphism of Lie algebra objects in $\mathbf{Gr}_{\leq N}(Q_+)$. Moreover,

$$\tau_{\leq N} U(\pi) : \tau_{\leq N} U(\mathfrak{n}_+(A)) \xrightarrow{\sim} \tau_{\leq N} U(L) \quad (11)$$

is an isomorphism of bialgebra objects in that category.

Proof. The assignment $e_i \mapsto x_i$ gives a surjection $\mathfrak{F} \twoheadrightarrow L$. The displayed relations place R_A in its kernel, so it factors through the unique surjection (9). For each β of height at most N , the map π_β is a surjection between finite-dimensional spaces of the same dimension and hence is an isomorphism.

Give both enveloping algebras the PBW filtration. The associated graded map of $U(\pi)$ is

$$\text{Sym}(\pi) : \text{Sym}(\mathfrak{n}_+(A)) \longrightarrow \text{Sym}(L).$$

In a Q_+ -degree β of height at most N , every symmetric monomial uses only Lie components of height at most N ; hence $\text{Sym}(\pi)_\beta$ is an isomorphism. The PBW filtration in a fixed positive height is finite, so $U(\pi)_\beta$ is an isomorphism. Functoriality of the standard primitive coproduct makes $U(\pi)$ a bialgebra map, and strong monoidality of $\tau_{\leq N}$ gives (11). $\square$

Remark 4.2. *The dimensions in (10) do not create a bracket. The relations and the chosen generators first produce the surjection (9); the dimensions only show that its kernel vanishes through height N .*

5 Associative recognition and reverse quotients

A recognized Lie algebra acts through its enveloping algebra. Its image in a Hall algebra can satisfy additional associative relations. The comparison must therefore identify that image and its kernel separately.

Let $\mathcal{H} = \bigoplus_{\beta \in Q_+} \mathcal{H}_\beta$ be a unital Q_+ -graded associative algebra, and let $x_i \in \mathcal{H}_{\alpha_i}$. Denote by $L \subset \mathcal{H}$ the Lie subalgebra generated by the x_i under the commutator, and by $\mathcal{H}^{\text{sph}} \subset \mathcal{H}$ the unital associative subalgebra generated by the same elements. The inclusion $L \hookrightarrow \mathcal{H}$ induces an algebra map

$$U(L) \longrightarrow \mathcal{H}^{\text{sph}},$$

but Lie recognition alone says nothing about its kernel.

Proposition 5.1 (Finite-height associative recognition). *If the x_i satisfy (6) and (7), there is a unique surjective graded algebra map*

$$\Phi : U(\mathfrak{n}_+(A)) \twoheadrightarrow \mathcal{H}^{\text{sph}}, \quad e_i \longmapsto x_i. \quad (12)$$

If

$$\dim_{\mathbb{k}} \mathcal{H}_\beta^{\text{sph}} = \dim_{\mathbb{k}} U(\mathfrak{n}_+(A))_\beta \quad (\text{ht}(\beta) \leq N), \quad (13)$$

then $\tau_{\leq N} \Phi$ is an isomorphism of associative algebra objects in $\mathbf{Gr}_{\leq N}(Q_+)$. If $\mathcal{H}$ is a graded bialgebra and the x_i are primitive, then $\mathcal{H}^{\text{sph}}$ is a graded sub-bialgebra, Φ is a bialgebra map, and the same conclusion holds for bialgebra objects.

Proof. The relations give a Lie map $\mathfrak{n}_+(A) \rightarrow \mathcal{H}$ with image L . Its enveloping map has image exactly the unital algebra generated by the x_i , proving (12). Under (13), every homogeneous component through height N is a surjection between finite-dimensional spaces of equal dimension. If the x_i are primitive, the standard coproduct on $U(\mathfrak{n}_+(A))$ and the coproduct on $\mathcal{H}$ agree on the algebra generators, which proves the bialgebra assertion. $\square$

For $N \geq 1$, put

$$B_N = \tau_{\leq N} U(\mathfrak{n}_+(A)), \quad S_N = \tau_{\leq N} \mathcal{H}^{\text{sph}}, \quad H_N = \tau_{\leq N} \mathcal{H},$$

and write $j_N : S_N \hookrightarrow H_N$ for the inclusion and $\Phi_N = \tau_{\leq N} \Phi$.

Proposition 5.2 (The reverse quotient is additional data). *Assume the hypotheses producing (12). If there is a unital graded algebra morphism in $\mathbf{Gr}_{\leq N}(Q_+)$*

$$\rho_N : H_N \longrightarrow B_N, \quad \rho_N(x_i) = e_i, \quad (14)$$

then ρ_N is surjective and

$$\rho_N j_N \Phi_N = \text{id}_{B_N}. \quad (15)$$

Consequently, $\Phi_N : B_N \rightarrow S_N$ is an isomorphism with inverse $\rho_N|_{S_N}$, and

$$H_N / \ker \rho_N \cong B_N.$$

The map ρ_N is an isomorphism precisely when $H_N = S_N$. In particular, this holds if all homogeneous components are finite-dimensional and

$$\dim_{\mathbb{k}}(H_N)_\beta = \dim_{\mathbb{k}}(B_N)_\beta \quad (\text{ht}(\beta) \leq N).$$

If H_N is a bialgebra object, the x_i are primitive, and ρ_N is a bialgebra morphism, then S_N is a sub-bialgebra, Φ_N and its inverse are bialgebra isomorphisms, and $H_N / \ker \rho_N \cong B_N$ as bialgebras.

Proof. The composite in (15) fixes the unit and every generator e_i of B_N , hence is the identity. It follows at once that ρ_N is surjective and that Φ_N is injective. Proposition 5.1 makes Φ_N surjective, so its left inverse $\rho_N|_{S_N}$ is its inverse. For $h \in H_N$,

$$h - j_N \Phi_N \rho_N(h) \in \ker \rho_N,$$

and $S_N \cap \ker \rho_N = 0$. Hence $\ker \rho_N = 0$ is equivalent to $H_N = S_N$. The dimension criterion makes each surjective homogeneous component of ρ_N injective. Under the final hypotheses, S_N is a sub-bialgebra and all the maps in (15) preserve the bialgebra operations. $\square$

If T_N is the free unital algebra object on the simple generators and $I_{B,N}$ and $I_{S,N}$ are the kernels of its maps to B_N and S_N , then Φ_N records the inclusion

$$I_{B,N} \subseteq I_{S,N}.$$

A reverse map fixing the generators requires the opposite inclusion. Thus neither (12) nor a scalar denominator constructs (14). PBW and a componentwise dimension equality can construct the inverse on S_N , but extending that inverse to H_N still requires generation or an independent splitting. On the spherical subalgebra, the existence of (14) forces $I_{B,N} = I_{S,N}$.

Remark 5.3 (Lie PBW is not Hall PBW). For $A = (0)$, take

$$\mathcal{H} = \mathbb{k}[\varepsilon]/(\varepsilon^2), \quad \deg \varepsilon = \alpha, \quad x = \varepsilon.$$

Then $L = \mathbb{k}\varepsilon \cong \mathfrak{n}_+(A)$, so Theorem 4.1 recognizes the Lie algebra in every height. Nevertheless,

$$U(\mathfrak{n}_+(A)) = \mathbb{k}[e] \longrightarrow \mathcal{H}, \quad e \longmapsto \varepsilon$$

has kernel (e^2) . Thus (10) cannot replace the associative dimension comparison (13). Moreover, for $N \geq 2$ there is no map (14) sending ε to e , since it would send $\varepsilon^2 = 0$ to the nonzero element $e^2 \in B_N$. To recognize all of $\mathcal{H}$, one must also prove $\mathcal{H} = \mathcal{H}^{\text{sp}}; this fails whenever, for example, $\mathcal{H}_0$ contains more than the unit.$

6 Compatible height completion

A comparison in every finite height determines a completed map only when the comparisons agree under truncation. The relevant topology and tensor product can be stated directly.

For a nonnegatively Q_+ -graded vector space V , set

$$F^{N+1}V = \bigoplus_{\text{ht}(\beta) > N} V_\beta, \quad \widehat{V} = \varprojlim_N V/F^{N+1}V = \prod_{\beta \in Q_+} V_\beta.$$

The transition maps are the projections in (8). The tensor product appropriate to this topology is

$$\widehat{V} \widehat{\otimes}_{\text{ht}} \widehat{W} = \prod_{\beta \in Q_+} \left(\bigoplus_{\gamma + \delta = \beta} V_\gamma \otimes W_\delta \right). \quad (16)$$

Each inner sum is finite because I is finite.

Proposition 6.1 (Formal height completion). *Let $f : V \rightarrow W$ be a graded linear map. If $\tau_{\leq N} f$ is an isomorphism for every N , then f is a componentwise isomorphism and extends to a continuous isomorphism*

$$\widehat{f} : \widehat{V} \xrightarrow{\sim} \widehat{W}.$$

If V and W are graded Lie algebras, associative algebras, or bialgebras and f is a morphism of the corresponding kind, then $\widehat{f}$ respects the continuous operations; for bialgebras the completed coproduct takes values in (16).

Proof. For a fixed β , any $N \geq \text{ht}(\beta)$ shows that f_β is an isomorphism. Taking the product of the maps f_β gives $\widehat{f}$. A graded product, bracket, or coproduct has only finitely many summands in each fixed total degree, so its continuous extension is defined componentwise and is preserved by $\widehat{f}$. $\square$

Corollary 6.2. *If the hypotheses of Theorem 4.1 and (10) hold in every degree, then*

$$\widehat{\mathfrak{n}_+(A)} \cong \widehat{L}, \quad U(\widehat{\mathfrak{n}_+(A)}) \cong \widehat{U(L)}$$

as complete Lie algebras and complete bialgebras, respectively. The second formula denotes the height completion of the enveloping algebra; it does not assert that the enveloping functor commutes with inverse limits.

Proposition 6.3 (Compatible reverse quotients). *Suppose that maps (14) exist for every N and, for $M \leq N$,*

$$\tau_{\leq M}(\rho_N) = \rho_M \quad (17)$$

under the identifications (8). Then inverse limits give a continuous split epimorphism

$$\widehat{\rho} : \widehat{\mathcal{H}} \twoheadrightarrow U(\widehat{\mathfrak{n}_+(A)})$$

whose compatible section is induced by the maps $j_N \Phi_N$, and

$$U(\widehat{\mathfrak{n}_+(A)}) \xrightarrow{\sim} \widehat{\mathcal{H}^{\text{sph}}}.$$

If every ρ_N is injective, the full completed map is an isomorphism. If $\mathcal{H}$ is a graded bialgebra, the x_i are primitive, and every ρ_N is a bialgebra morphism, these conclusions respect the continuous bialgebra operations and the completed tensor product (16).

Proof. Equation (17) defines $\hat{\rho}$ componentwise. The sections $j_N \Phi_N$ are compatible and satisfy (15), so their inverse limit is a section; no general preservation of epimorphisms by inverse limits is needed. Proposition 5.2 identifies every spherical window, and Proposition 6.1 extends the operations continuously. If every ρ_N is injective, every finite window and hence the inverse limit is identified. $\square$

This formal conclusion applies to a single graded comparison, or to a compatible inverse system already known to arise from such graded objects. It does not identify an independently defined topological or derived Hall carrier with that inverse limit.

7 Compact realization and amplitude comparisons

The algebraic results identify the maps needed from the geometric construction. Relations first give the enveloping-to-spherical map Φ . A reverse quotient ρ_N is additional data. Its kernel measures the part of the finite Hall window outside the recognized spherical subalgebra.

For the translation quotient, the coefficient realization must first apply to (4) and its unit BE . For the chosen reduced complexes, establish a determinant identity and coherent square-root maps analogous to (5), including their relation to the unreduced cross line and the corresponding flag coherences. The following comparison problem includes this quotient case only after those data have been constructed.

Open Problem 7.1 (Compact Hall–Borcherds comparison). *For a smooth projective Calabi–Yau threefold and a specified extension-closed sector, construct strong orientation data and the associative product (3) in an explicit charge-twisted monoidal realization, together with a strong monoidal totalization to the ordinary Q_+ -graded vector spaces of Section 5. In the resulting degree-zero associative algebra, choose a finite multiplicity-one Borcherds datum A and classes $x_i \in \mathcal{H}_{\alpha_i}$ of even cohomological parity. With the ordinary commutator $[x, y] = xy - yx$, prove the relations (6)–(7). This must first produce the surjection*

$$\Phi : U(\mathfrak{n}_+(A)) \twoheadrightarrow \mathcal{H}^{\text{sph}}.$$

Independently construct a Hall coproduct for which the x_i are primitive and, for every N , a graded bialgebra epimorphism

$$\rho_N : \tau_{\leq N} \mathcal{H} \twoheadrightarrow \tau_{\leq N} U(\mathfrak{n}_+(A)), \quad \rho_N(x_i) = e_i, \quad (18)$$

compatible as in (17). Proposition 5.2 then identifies every spherical window without a dimension calculation. Determine $\ker \rho_N$. If the intended target is the full Hall algebra, prove $\ker \rho_N = 0$ in every window; by Proposition 5.2, this is equivalent to $\tau_{\leq N} \mathcal{H} = \tau_{\leq N} \mathcal{H}^{\text{sph}}$. Componentwise finite-dimensional equality with B_N is one possible proof of this last step. Neither the scalar denominator nor the opposite-direction map Φ_N constructs (18).

For a completed comparison, specify a separated complete Hall topology, continuous product and coproduct, transition morphisms realizing every finite window, and an identification of its completed tensor product with (16). The compatible sections in Proposition 6.3, rather than a general exactness claim for inverse limits, then give the completed split epimorphism. If the carrier is derived, also identify it with the derived inverse limit and justify the disappearance of higher inverse limits. These requirements are not supplied by a scalar denominator. For $X = S \times E$ modulo translation, construct these coefficients and operations on the common-torsor

quotient over BE . Specify the realization of its unit and its relative tensor product. Any passage to ordinary graded vector spaces must preserve these operations. The absolute affine-stabilizer theorem does not supply this passage.

A representation introduces a further map

$$r : \mathcal{H} \longrightarrow \text{End}(V),$$

where the space V , its grading, and any completion must be specified. The root operator is then $r(x_i)$. Its algebra relations follow from a constructed algebra map r . A scalar generating series does not specify this representation.

An amplitude also requires a trace or pairing on a declared domain. For example, fix a subalgebra $\mathcal{D} \subseteq \text{End}(V)$ and a linear map $\lambda : \mathcal{D} \rightarrow \Lambda$ to a one-dimensional value space. If λ is a trace, it must satisfy $\lambda(ab) = \lambda(ba)$ on its domain. For a graded trace, the corresponding Koszul sign is required. The amplitude of an allowed Hall element h is $\lambda(r(h))$. Infinite sums require a topology in which both the representation and the trace are defined.

For an automorphic comparison in a parameter family T , the Hall construction must produce a section s_H of a line bundle $\mathcal{L}_H$. An independently defined automorphic section s_{aut} belongs to its own line bundle $\mathcal{L}_{\text{aut}}$. The required comparison is an isomorphism

$$\eta : \mathcal{L}_H \xrightarrow{\sim} \mathcal{L}_{\text{aut}}, \quad \eta(s_H) = s_{\text{aut}},$$

with the specified group actions and normalization. Finite-height recognition supplies neither this line isomorphism nor the trace that defines s_H . It supplies the algebraic comparison to which those additional constructions must be attached.

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